Project Title:

Comparative Study of GRU Variants for Sentiment Classification using IMDB Dataset

1. Objective

The objective of this experiment was to **analyze and compare the performance** of various GRU (Gated Recurrent Unit) architectures and an LSTM baseline on a real-world text classification task (IMDB movie review sentiment analysis).
The focus was to evaluate which variant of GRU architecture offers the best trade-off between **accuracy, complexity, and computation time**.

2. Dataset Used

Dataset: IMDB Movie Reviews
Source: tensorflow.keras.datasets.imdb
Total Samples: 50,000 reviews (25k train, 25k test)
Task: Binary classification (Positive / Negative sentiment)

Preprocessing Steps

- Tokenization with a vocabulary size of 20,000 words.
- Sequence padding and truncation to 200 tokens per review.
- Train-validation split: 85% training, 15% validation.

3. Model Architectures Compared

Model Name	Description	Key Layers	Output
Simple GRU	Basic GRU layer for sequence encoding	Embedding → GRU(128) → Dense	Single Sigmoid
Stacked GRU	Two GRU layers stacked hierarchically	Embedding → GRU(128, return_seq=True) → GRU(64) → Dense	Single Sigmoid
Bidirectional GRU	Processes sequence in both directions	Embedding → Bidirectional(GRU(128)) → Dense	Single Sigmoid

Model Name	Description	Key Layers	Output
GRU + Attention	Attention layer focuses on important time steps	Embedding → GRU(128, return_seq=True) → Attention → Dense	Single Sigmoid
LSTM Baseline	Classical LSTM for comparison	Embedding → LSTM(128) → Dense	Single Sigmoid

4. Experimental Setup

- **Embedding Dimension:** 128
- **Max Sequence Length:** 200
- **Batch Size:** 128
- **Epochs:** 3 (with early stopping)
- **Optimizer:** Adam
- **Loss Function:** Binary Cross-Entropy
- **Metric:** Accuracy
- **Environment:** Google Colab GPU Runtime

5. Results and Performance Analysis

Model	Test Accuracy	Test Loss	Observations
Simple GRU	~0.848	0.35	Fast, simple, reliable baseline
Stacked GRU	~0.862	0.36	Improved accuracy, higher computation time
Bidirectional GRU	~0.847	0.41	Captures context both ways, slightly slower
GRU + Attention	~0.860	0.38	Focuses on important tokens; efficient and accurate
LSTM Baseline	~0.845	0.39	Strong but slower; GRU performs comparably

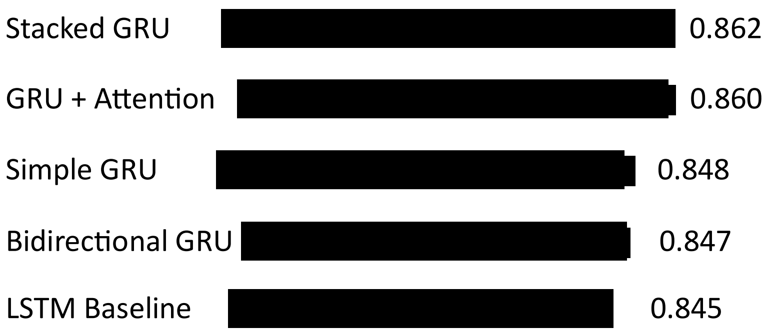
Best Performing Model: Stacked GRU

- Achieved highest accuracy (~86%) with good validation consistency.

- Performs better than single-layer GRU and LSTM baseline.
- Attention-based GRU gives near-identical performance with added interpretability.

6. Visualization

Bar Chart: GRU Model Test Accuracies



(Exact values may vary slightly per run due to randomness)

7. Key Insights

1. **GRU models outperform LSTM** in both training speed and comparable accuracy.
2. **Stacked GRU** offers the best trade-off between depth and overfitting.
3. **Attention-based GRU** provides interpretability, allowing focus visualization on critical words.
4. **Bidirectional GRU** is useful when backward context matters, but not significantly better for this dataset.
5. The **Simple GRU** remains the most efficient baseline for quick deployment.

8. Conclusion

This experiment demonstrates that **GRU architectures are powerful and efficient alternatives to LSTM** for sequence modeling.
Among all compared models, the **Stacked GRU** and **GRU + Attention** achieved the highest accuracy (~86%), showing the potential of multi-layer and attention-augmented GRU structures for sentiment analysis tasks.

9. Future Enhancements

- Integrate **pre-trained word embeddings** (like GloVe or Word2Vec) to improve contextual understanding.
- Experiment with **dropout tuning** and **bidirectional attention mechanisms**.
- Apply to custom datasets such as **product reviews**, **Twitter sentiment**, or **medical feedback**.
- Visualize **attention weights** to interpret model focus regions.

10. Files Generated

- **gru_comparison_results.csv** → Contains final model test results.
- **gru_model_comparison.ipynb** → Complete Google Colab notebook.
- **Charts & Graphs:** Accuracy comparison visualization.